# Almira Raisa Izzatina

Jakarta, Indonesia • +6287851080731
almirari004@gmail.com • linkedin.com/in/almira-ri • github.com/almirari/ • almira-ri.vercel.app

Data-driven problem solver with experience translating technical analysis into business decisions, cross-functional collaboration, and stakeholder communication.

## Experiences

**Accenture Southeast Asia** - Central Jakarta, Jakarta
***Technology & Consulting Intern*** **Jan 2026 - Sep 2026**

- Validated production reliability across 3 interconnected microservice domains (order fulfillment & provisioning, an integration/middleware layer handling request routing and payload transformation, and an orchestration layer between the sales/channel and fulfillment systems) supporting a telecommunications provider serving 50M+ subscribers, within an 880+ issue delivery pipeline, catching 20+ critical pre-production defects via systematic API contract testing against JSON schema specs.
- Independently diagnosed root-cause failures in containerized services by parsing logs in OpenShift and correlating live test results against API specifications.
- Ensured data integrity across fulfillment and account management workflows using SQL-based validation, coordinating defect resolution under Agile sprint cadence.

**PT. Bank Syariah Indonesia** - Central Jakarta, Jakarta
***Data Analyst Intern – Collections Strategy & Performance Team*** **Jul 2025 - Sep 2025**

- Automated cleaning and feature extraction across 2M+ repayment records in Python, cutting processing time by 70% and surfacing delinquency patterns that directly informed collections strategy decisions.
- Built a repayment behavior analysis pipeline correlating contact frequency with payment outcomes, producing contact-timing recommendations that contributed to a reported 12% improvement in collections efficiency.

**Intelligent Computation & Vision Laboratory** - Sepuluh Nopember Institute of Technology, Surabaya
***Assistant Researcher*** **Feb 2025 - Present**

- Published and presented a Teacher–Student Knowledge Distillation framework at ICTS 2025 (DOI:10.1109/ICTS67612.2025.11369518) that compressed a vision model to 2.23M parameters while maintaining 95.64% accuracy, enabling low-power, real-time deployment for agricultural field environments with no GPU access.
- Designed and benchmarked efficient neural architectures in PyTorch, evaluating accuracy–latency–memory tradeoffs against SOTA baselines to identify optimal compression configurations for edge deployment.

**Intelligent Computation & Vision Laboratory** - Sepuluh Nopember Institute of Technology, Surabaya
***Laboratory Administrator*** **May 2024 - Aug 2026**

- Organized introductory machine learning workshops for freshmen, covering essential topics such as regression, classification, clustering, and their practical applications. Utilized hands-on programming exercises to reinforce understanding and foster technical skills in students.

## Education

**Institut Teknologi Sepuluh Nopember** - Surabaya, Indonesia
Bachelor of Informatics Engineering, 3.61/4.00 **Jul 2022 - Aug 2026**

**Research Interests:** Resource-Efficient AI, Computer Vision for Sustainability, AI for Social Impact.

## Recent Projects

**TanyaRasa: Indonesian Mental Health Chatbot with Fine-Tuned IndoBERT** **PREVIEW**
Built an end-to-end NLP intent-classification pipeline for an Indonesian-language mental health chatbot: collected, translated, cleaned, and labeled conversational text data, then fine-tuned IndoBERT for intent classification, achieving 85.35% accuracy and 84.59% F1-score. Benchmarked against a generative baseline (CENDOL) and deployed the model via Streamlit and Hugging Face.

**Digital Lung Stethoscope with CNN and Knowledge Distillation for Respiratory Disease Detection** **PREVIEW**
Engineered an end-to-end signal-processing and deep learning pipeline deployed on Raspberry Pi hardware, enabling low-cost real-time diagnosis in resource-limited clinics. This work was recognized with an Honorable Mention at GEMASTIK XVIII 2025 (The Ministry of Education, Culture, Research and Technology) for scientific writing and innovation.

**Knowledge Distillation for Efficient Ordinal Classification of Oil Palm Ripeness Level** **PREVIEW**
Built a compressed AI model to optimize harvest timing decisions, supporting sustainable agricultural practices with field-deployable, low-power hardware.

## Technical Skills

**Analytics & Data -** Python, SQL, Excel, Scikit-learn, Statistical Modeling, Data Cleaning & Pipeline Automation, TensorFlow, PyTorch, HuggingFace

**Professional -** Cross-functional collaboration, Agile delivery, stakeholder communication, problem structuring.

## Achievements

**Honorable Mention of ICT Scientific Writing Division at GEMASTIK XVIII 2025 - Top 4 of 386 teams**
*Issued by The Ministry of Education, Culture, Research and Technology (MoECRT)* **Oct 2025**
Awarded for the innovation of the "Digital Lung Stethoscope with CNN and Knowledge Distillation for Respiratory Disease Detection" research.

**3rd Place of Presentation Event Formula SAE Japan 2024 - Top 3 of 75 teams**
*Issued by Formula Society of Automotive Engineers* **Sep 2024**
Translated financial metrics and marketing strategy into visuals for an international panel presentation.

**Top 10 Team at VWO UI 2023 UI/UX Competition**
*Issued by University of Indonesia Vocational Student Executive Board* **Nov 2023**
Led user research and interface design for a regional language-learning app.

**3rd Place of Presentation Event Formula SAE Japan 2023 - Top 3 of 77 teams**
*Issued by Formula Society of Automotive Engineers* **Sep 2023**
Designed a business plan presentation using data storytelling to communicate key insights using digital transformation strategies.

**Most Innovative Paper at Youth Idea Competition NBRI 2023**
*Issued by National Battery Research Institute* **Aug 2023**
Conducted predictive analysis of EV charging infrastructure in Indonesia, optimizing charging station placement based on user behavior and energy demand patterns.

**1st Place of Overall Event of Formula Bharat Pi-EV**
*Issued by Formula Bharat India* **Jul 2023**
Transformed EV engineering data into an award-winning presentation.